

CRISPR-PLANT gRNA Designer with Contextual Intelligence and AI-Powered Decision Support: Automated Off-Target Analysis, Safety Prioritization, and Gene Function Insights for Plant Genome Editing

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Abstract

Summary

Here we present **CRISPR-PLANT gRNA Designer**, a web application automates guide RNA (gRNA) selection for plant genome editing by combining the CRISPR-PLANT server, a safety-prioritized multi-criteria ranking algorithm, and two contextual intelligence layers: OpenAlex literature integration for off-target genes and Google Gemini AI assistant with detailed system prompts understanding the user's target gene, top candidates, and critical genes. The tool eliminates manual screening, provides personalized gRNA recommendations grounded in specific analysis data, and enables rules and AI guided decision-making on trade-offs between efficiency and genomic safety. By integrating robust automation, literature context, and AI-powered decision support, the platform reduces manual effort and improves the likelihood of successful precise genome editing in plant systems.

Availability and Implementation

The application is freely available as a web app at <https://bioinfo.icgeb.res.in/sharp>

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Supplementary information

Supplementary materials are available at *Bioinformatics* online.

1. Introduction

The growing global population, projected to reach 9.7 billion by 2050 (United Nations, 2019) places unprecedented pressure on agricultural systems to enhance crop productivity, nutritional quality, and resilience against biotic and abiotic stresses. Traditional breeding methods, while successful, are often time-consuming and limited by existing genetic variation. In this context, the CRISPR toolbox has revolutionized plant science by enabling programmable, transgene-free genome editing for crop improvement, where ribonucleoprotein (RNP) delivery eliminates foreign DNA integration while allowing precise targeted modifications to develop new germplasm and address agricultural challenges in the face of climate change and population growth

(Zhang et al., 2019). This technology has been successfully applied in major crops including rice, wheat, maize, and soybean for traits such as disease resistance, drought tolerance, and yield enhancement (Zafar et al., 2020). A critical determinant of CRISPR-Cas9 editing success is the selection of effective and specific single-guide RNAs (sgRNAs). For brevity we will use the term gRNA for single guide RNA abbreviation as "gRNA". Optimal gRNA design requires balancing multiple competing criteria: high on-target activity typically predicted by scoring algorithms like Rule Set 2 or CFD (Doench et al., 2016; Hsu et al., 2013), optimal GC-content in the PAM-distal region (>25% GC within nucleotides 4-13 of the sgRNA), presence of purine nucleotides at the PAM-proximal position 20, and avoidance of predicted off-target sites (Labuhn et al., 2018). Off-target editing remains a primary concern in complex crop genomes, particularly in polyploid species where high

Table 1. Comparison with other tools.

Tool Name	Cross-gRNA Ranking	Scoring Algorithms Used	Specialized Features
CRISPR-PLANT gRNA Designer:	Yes (safety-first, multi-criteria ranking guides)	CFD (via backend)	Intelligent ranking, Research paper database, LLM integration
CRISPR-P 2.0	No (outputs raw list)	Rule Set 1/2, CFD	75 plant genomes
CRISPR-GE	No (outputs raw list)	CFD	40+ plant species, Primer design
CRISPR-PLANT v2	No	Not specified	Model plant focus
CRISPOR	No (Per guide ranking)	10+ algorithms	Flexible algorithm selection
CHOPCHOP	No (Per guide ranking)	Rule Set 2	Repair outcome prediction
CRISPR RGEN Tools	No (Per guide ranking)	Multiple (Cas-OFFinder)	Base & Prime editing support
WGE (Sanger)	No	Mismatch-based only	Mismatch-only analysis
CRISPR-Cereal	No (outputs raw list)	Rule Set 2	Cereal crop focus
DeepCRISPR	No (Per guide ranking)	Deep Learning	End-to-end deep learning
CRISPRon	No (Per guide ranking)	Transformer	Epigenome integration

sequence homology between subgenomes increases the risk of unintended modifications, and in crops like maize where approximately 85% of the genome consists of highly repetitive sequences that create numerous closely matched sites, complicating specificity assessment and off-target detection in functionally important regions (Lee et al., 2018). The process of selecting optimal guides remains complex and labor-intensive. Researchers navigate multiple tools each with distinct interfaces and metrics, a process that can consume hours to days per target gene. Just for the case of knockdown of a single gene there were 176 possible gRNAs, each with multiple off targets. The total off targets were 3000+ making it a daunting task to assess all off targets manually.

Current plant-specific gRNA design platforms have made significant contributions but crucially none of these tools performs safety-first re-ranking that jointly minimizes off-target burden across functional genomic regions, nor do they link off-target predictions to biological literature. CRISPR-P 2.0 (Liu et al., 2017) supports 75+ plant species but lacks integrated ranking systems. CRISPR-GE (Xie et al., 2017) provides genome-specific prediction but requires manual filtering of results. While CRISPOR (Haeussler et al., 2016) incorporates multiple scoring algorithms, its plant genome support is secondary to mammalian systems. Advanced deep-learning tools such as DeepCRISPR (Chuai et al., 2018) and CRISPRon (Xiang et al., 2021) have improved on-target efficiency prediction but were trained exclusively on human cell line data and offer no support for plant genomes. The CRISPRon/off interface (Anthon et al., 2022) similarly covers only six animal species. Table 1 highlights these comparative limitations across 10 established tools, revealing a consistent gap in automated, multi-criteria analysis pipelines specifically optimized for plant genomes.

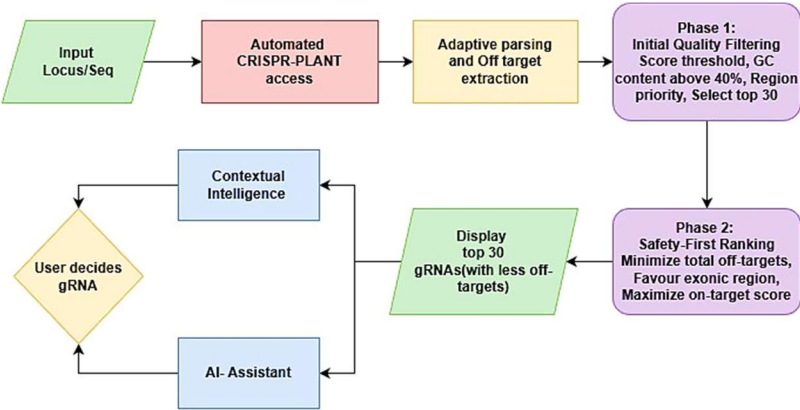
To address these challenges, we present an automated CRISPR gRNA Design and Analysis Pipeline that integrates intelligent parsing, multi-criteria ranking, and contextual analysis into a unified workflow. Our system builds upon the established CRISPR-PLANT tool (Liu et al., 2017) but extends its functionality through adaptive web scraping algorithms that reliably extract and interpret results across variable website structures.

The pipeline implements a novel two-phase prioritization system that first filters gRNAs by established quality thresholds (score high to low, GC content > 40%), then ranks candidates using a weighted scoring system that balances on-target efficiency, genomic context (prioritizing exonic regions), and off-target risk with particular emphasis on off-targets occurring in coding sequences and regulatory regions. Unique to our platform is the integration of an AI-powered literature assistant that queries OpenAlex (Priem et al., 2022) database to retrieve relevant publications for both target genes and critical off-target genes.

This contextual intelligence transforms off target lists from abstract predictions into biologically meaningful risks, enabling researchers to make informed decisions based on existing knowledge about gene functions and potential pleiotropic effects. Furthermore, the integration of a conversational AI assistant (powered by Google's Gemini API) offers real-time guidance through natural language interaction welcoming researchers across experience levels to ask questions about results, explore biological contexts, and receive immediate, context-aware insights.

Our tool reduces gRNA design from days to minutes through automated, context-aware analysis, accelerating crop improvement pipelines and shifting plant genome editing toward intelligent, end-to-end solutions.

A)



B)

1. Input Parameters

Input Type

☒ Locus Tag

☐ Sequence

☐ Genomic Position

Locus Tag

GLYMA14G07880

PAM Sequence

NGG (SpCas9)

Guide Sequence Length

20 bp

snoRNA Promoter

☒ U3 (default)

☐ U6

2. Genome Selection

Fetch Available Genomes

CRISPR gRNA Design & Analysis Pipeline

This tool automates gRNA design using the CRISPR-PLANT website, then performs a comprehensive off-target analysis to identify the safest and most effective candidates.

Analysis Results

Found and prioritized 30 high-quality gRNAs based on: on-target score, GC content, genomic region, and off-target analysis.

Rank	Sequence	GC Content	Region	Critical Off-Targets	Total Off-Targets	Critical Genes
2	TTTGACTGCGATCCATCTCTGG	45.0%	exon	2	13	GLYMA13G05068, GLYMA19G02260
3	CAGCTGACTGGGTTTACAGAGGG	50.0%	exon	2	20	GLYMA19G03401, GLYMA13G05890
4	ATCACAATGTGTCATCAACCTTGG	45.0%	exon	2	20	GLYMA12G37000, GLYMA19G28541
5	ATACGTAGGCTCCAAGGAATGG	45.0%	exon	2	20	GLYMA06G03310, GLYMA04G03240
6	GACCAGGGAATCTCTGAGTTGG	50.0%	exon	2	20	GLYMA06G03350, GLYMA11G08630

Ask about your gRNA results...

C)

Guide Sequence Length

20 bp

snoRNA Promoter

☒ U3 (default)

☐ U6

2. Genome Selection

Fetch Available Genomes

Select Genome

Glycine max (V1.0)

82 genomes loaded successfully!

3. Run Analysis

Design and Analyze gRNAs

AI Assistant - Ask Questions About Your Results

Ask me about your gRNA results, off-target effects, or CRISPR design strategies!

What makes a good gRNA for CRISPR experiments?

Designing an effective gRNA for CRISPR/Cas9 experiments hinges on achieving high on-target cleavage efficiency while minimizing unintended off-target edits. A "good" gRNA, therefore, ideally possesses a high on-target specificity score, optimal GC content, and crucially, the fewest possible critical off-target sites. The on-target score predicts how efficiently the gRNA will guide Cas9 to the desired genomic location, with higher scores generally correlating with better editing success. An ideal GC content, typically between 40-60%, ensures proper gRNA-DNA duplex stability, which is vital for robust Cas9 activity. Furthermore, targeting an early exon within the gene is generally preferred for knockout experiments, as it maximizes the chances of inducing a frameshift mutation that leads to a non-functional protein.

However, the most critical consideration for gRNA selection is the assessment of off-target activity. In the provided analysis, the number of "Critical Off-targets" and the list of "Critical Genes" are vital. A gRNA with fewer critical off-targets is highly desirable, as it reduces the risk of undesirable mutations in other parts of the genome. Given the complexity of the *Glycine max* genome, with its history of polyploidy and duplicated genes, it is imperative for the user to carefully

Ask about your gRNA results...

D)

Guide Sequence Length

20 bp

snoRNA Promoter

☒ U3 (default)

☐ U6

2. Genome Selection

Fetch Available Genomes

Select Genome

Glycine max (V1.0)

82 genomes loaded successfully!

3. Run Analysis

Design and Analyze gRNAs

Critical Off-Target Genes Research

Explore research papers for each gene identified in your critical off-target analysis.

Found 128 unique critical genes across all gRNAs

Papers for gene: GLYMA06G03310

Found 4 papers for GLYMA06G03310

- 1. Expression profiling and in silico homology modeling of inositol pentakisphosphate 2-kinase, a potential candidate gene for low phytate trait in soybean (2020)
- 2. Mutation of GmIPK1 Gene Using CRISPR/Cas9 Reduced Phytic Acid Content in Soybean Seeds (2022)
- 3. Phytic acid phosphorus and inorganic phosphorus composition in seeds of soybean lines with independent inositol pentakisphosphate 2-kinase mutations (2013)

Ask about your gRNA results...

Fig.1 A) Workflow B) User Interface and result section C) AI Assistant D) Critical genes research.

2. Materials and Methods

2.1 Tool Architecture and Implementation

The pipeline was implemented as a Streamlit web application using Selenium WebDriver for headless automation of the CRISPR-PLANT v2 server (Liu et al., 2017), with a modular architecture spanning automated submission, results parsing, off-target extraction, and intelligent ranking (Fig. 1A).

The absence of a public API necessitated web automation techniques, while variable server response times required adaptive timeout mechanisms to accommodate network latency and processing delays.

2.2 Automated Server Navigation and Job Submission

The pipeline initiates launching a headless Chrome browser instance configured with extended timeouts to handle the CRISPR-PLANT server's variable response times. Upon user input (locus tag, sequence, or genomic position), the tool navigates to the CRISPR-PLANT website and traverses to the gRNA design form through a "Start" link. The form is populated programmatically with user-specified parameters including: (1) target genome selection from 75+ available plant genomes, (2) PAM sequence selection (NGG for SpCas9, NAG, NGA, NNGRRRT for SaCas9, TTTN for Cpf1 variants), (3) guide RNA length (15-22 bp), (4) snoRNA promoter selection (U3 or U6), and (5) the target input (locus identifier, DNA sequence, or genomic coordinates). The PAM and guide length selections are mapped to appropriate indices in the web form's dropdown menus using a predefined mapping dictionary based on CRISPR-PLANT's interface structure.

The tool handles three input modalities: locus tags (e.g., GLYMA14G07880), raw DNA sequences (20-2000 bp), or genomic coordinates (Chr1:12345-12500 format). Input validation ensures only one modality is active per submission. Once all parameters are configured, the form is submitted, and the system enters a waiting state for server processing, periodically checking for result table emergence.

2.3 Robust Results Parsing with Adaptive Fallback Mechanisms

To ensure reliable data extraction from the dynamic CRISPR-PLANT results page, we developed a three-tier parsing strategy that adapts to variations in page structure and element organization. This approach first applies semantic content analysis by scanning HTML tables for biological keywords ("Guide," "Score," "Sequence," "Region," "GC %"). If unsuccessful, it employs structural heuristics based on table size and data density. As a final fallback, minimal structural validation selects tables meeting basic dimensional requirements for gRNA data representation. This adaptive system successfully identified the target results table. For each validated table, we extracted gRNA sequences, on-target efficiency scores (0-1 range, representing CFD scores), GC content percentages, and genomic region annotations (exon, intron, UTR, CDS), with validation for nucleotide composition and length constraints. Detailed parsing algorithms and validation metrics are provided in Supplementary Methods.

2.4 Off-Target Data Extraction from CRISPR-P 2.0

The automation uses Selenium's ActionChains to trigger off-target displays and dynamically maps columns by analyzing header content, ensuring robustness against changes in column order or naming conventions. Each gRNA's predicted off-targets were annotated with genomic features, and sites falling within exons or other coding regions of non-target genes were treated as critical, as such off-target cutting has been shown to reduce cell fitness and confound growth phenotypes in CRISPR-Cas9 screens (Tycko et al., 2019). Non-critical off-targets in intronic or intergenic regions are recorded but weighted less heavily in subsequent ranking.

2.5 Intelligent Multi-Criteria gRNA Ranking Algorithm

The pipeline implements a hierarchical ranking system that optimizes for both efficacy and specificity based on established best practices in gRNA design (Doench et al., 2016; Wilson et al., 2018; Labuhn et al., 2018). The algorithm proceeds through two phases:

Phase 1: Initial Quality Filtering and Pre-ranking

- Filter candidate gRNAs by on-target score (high to low).
- Filter by GC content > 40% (optimal for stable gRNA:DNA hybridization).
- Assign region priority scores: exon = 3, CDS/UTR = 2, intron = 1.
- Sort by region priority (descending), then by on-target score (descending).
- Select top 30 candidates for detailed off-target analysis.

Phase 2: Safety-First Final Ranking

Following off-target analysis, candidates undergo final prioritization using a four-level hierarchical sort:

- Primary: Critical off-target count (ascending - minimize functional off-targets).
- Secondary: Total off-target count (ascending - minimize overall off-target burden).
- Tertiary: Region priority (descending - favor exonic targeting).
- Quaternary: On-target score (descending - maximize efficiency among equally safe options).

Region priority scores in Phase 1 are assigned based on the genomic region of the guide RNA's own target site within the gene of interest, not off-target loci. For off-target analysis in Phase 2, hits are classified as 'critical' if they map to exon, CDS, or UTR regions of non-target genes; the critical off-target count is simply the total number of such functionally disruptive hits per gRNA, no weighted aggregation is applied. This safety-first hierarchy reflects the principle that minimizing unintended genomic modifications takes precedence over maximizing cutting efficiency, particularly in crop improvement applications where off-target effects could compromise plant viability or introduce unintended traits.

2.6 Literature Integration and AI-Assisted Analysis

The tool queries OpenAlex (Priem et al., 2022) for publications on critical off-target genes and integrates Google's Gemini 2.5 Flash as an interactive AI assistant that provides evidence-based gRNA

selection recommendations, answers user questions about analysis results and off-target risks and addresses broader CRISPR-Cas9 queries.

2.7 Tool comparison and user interface

We compared 10 publicly available CRISPR gRNA design tools (Table 1) across 3 functional dimensions and across 13 functional dimensions (in supplementary table), benchmarking against CRISPR-P 2.0 (Liu et al., 2017), CRISPR-GE (Xie et al., 2017), and CRISPOR (Haeussler et al., 2016). The Streamlit interface (Fig. 1B) features parameter configuration, interactive visualization (Fig. 1C), expandable off-target details, AI assistant (Fig. 1C), and literature search support (Fig. 1D).

3. Discussion and Conclusion

Our automated pipeline streamlines CRISPR gRNA design by combining server automation, adaptive parsing, and AI assistance to reduce design time from days to minutes. The tool provides biologically informed rankings with safety-first prioritization and literature-integrated off-target analysis critical advances for polyploid crops where specificity risks are highest.

Unlike deep learning approaches such as DeepCRISPR and CRISPRon, which rely on epigenetic training data available exclusively for human cell lines (Chuai et al., 2018; Xiang et al., 2021), our tool is built on plant-specific genome infrastructure and prioritizes biological safety over raw efficiency scores a distinction critical for crop improvement applications where off-target edits can compromise plant viability and regulatory compliance.

This integrated approach enables informed gRNA selection without requiring deep bioinformatics expertise, democratizing access to sophisticated analysis for crop improvement programs. However, as with all LLM-based systems, users should treat AI outputs as decision-support guidance rather than definitive recommendations and should verify biological interpretations against primary literature. Future development will incorporate chromatin accessibility data and expand to base-editing and prime-editing applications.

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Conflict of Interest: none declared.

References

Anthon C, Corsi GI, Gorodkin J. 2022; *Bioinformatics* 38:5437–39.

Chuai G, Ma H, Yan J, Chen M, Hong N, Xue D et al. 2018; *Genome Biol.* 19:80.

Doench JG, Fusi N, Sullender M, Hegde M, Vaimberg EW, Donovan KF et al. 2016; *Nat. Biotechnol.* 34:184–91.

Haeussler M, Schönig K, Eckert H, Eschstruth A, Mianné J, Renaud JB et al. 2016; *Genome Biol.* 17:1–12.

Hsu PD, Scott DA, Weinstein JA, Ran FA, Konermann S, Agarwala V et al. 2013; *Nat. Biotechnol.* 31:827–32.

Labuhn M, Adams FF, Ng M, Knoess S, Schambach A, Charpentier E et al. 2018; *Nucleic Acids Res.* 46:1375–85.

Lee K, Zhang Y, Kleinstiver BP, Guo JA, Aryee MJ, Miller J et al. 2019; *Plant Biotechnol. J.* 17:362–72.

Liu H, Ding Y, Zhou Y, Jin W, Xie K, Chen LL. 2017; *Mol. Plant* 10:530–32.

Priem J, Piwowar H, Orr R. 2022; *arXiv preprint arXiv:2205.01833*.

Tycko J, Wainberg M, Marinov GK, Ursu O, Hess GT, Ego BK et al. 2019; *Nat. Commun.* 10:4063.

United Nations Department of Economic and Social Affairs, Population Division. 2019; *World Population Prospects 2019: Highlights*. New York: United Nations.

Wilson LO, Reti D, O'Brien AR, Dunne RA, Bauer DC. 2018; *CRISPR J.* 1:182–90.

Xiang X, Corsi GI, Anthon C, Qu K, Pan X, Liang X et al. 2021; *Nat. Commun.* 12:3238.

Xie X, Ma X, Zhu Q, Zeng D, Li G, Liu YG. 2017; *Mol. Plant* 10:1246–49.

Zafar SA, Zaidi SSEA, Gaba Y, Singla-Pareek SL, Dhankher OP, Li X et al. 2020; *J. Exp. Bot.* 71:470–79.

Zhang Y, Malzahn AA, Sretenovic S, Qi Y. 2019; *Nat. Plants* 5:778–94.